

Daily Progress Report

Week 5_Day1: Component Architecture & JSX

Program	3-Month MERN Stack Internship Program
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Week	Week_5 (Day 1)
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Day 1: Component Architecture & JSX

1.1 Topics Studied

- React fundamentals and the component-based architecture
- Setting up a React development environment using **Vite**
- Comparison between **Vite** and **Create React App (CRA)**
- JSX (JavaScript XML) syntax and its role in building user interfaces
- Rendering elements and component trees in React
- Creating reusable Functional Components
- Understanding the differences between Functional and Class Components
- Passing data between components using **props**
- Organizing React applications through modular component architecture

1.2 Task Description

Developed a static React landing page by bootstrapping a new project using **Vite** and organizing the user interface into reusable components. The application was structured into separate **Navbar**, **Hero**, and **Footer** components to demonstrate React's component-based architecture. Basic textual content was passed from the parent component to child components using **props**, establishing a clear understanding of one-way data flow and component communication in React.

1.3 Implementation Details

- Initialized a new React project using Vite, providing a lightweight and high-performance development environment.
- Explored the default React project structure generated by Vite, including the purpose of main.jsx, App.jsx, and the components directory.
- Created a dedicated Navbar component responsible for displaying the website title.
- Developed a Hero component containing a heading, descriptive text, and a static Get Started button.
- Implemented a Footer component to display copyright information.
- Organized the application into reusable and independent React components following modern component-based development practices.
- Passed dynamic string values from the parent App.jsx component to child components using props, avoiding hardcoded content inside the components.
- Utilized JSX syntax to define the user interface with HTML-like elements directly within JavaScript.
- Rendered all components within the root App component to construct a complete static landing page.
- Applied basic CSS styling to create a clean, responsive layout with clearly separated navigation, hero, and footer sections.
- Maintained a modular project structure that promotes scalability, readability, and ease of future enhancement.

1.4 Outcome

Successfully developed a static React landing page that demonstrates the foundational concepts of React development, including project initialization with **Vite**, component-based architecture, JSX syntax, and component communication through **props**. The project strengthened my understanding of organizing user interfaces into reusable components while reinforcing React's one-way data flow model. Additionally, working with Vite provided practical experience with a modern React development environment, establishing a strong foundation for building scalable and maintainable React applications in future projects.

Screenshots:

```
src > App.jsx > default
1  import Navbar from "../components/Navbar";
2  import Hero from "../components/Hero";
3  import Footer from "../components/Footer";
4  import "../App.css";
5
6  function App() {
7    return (
8      <>
9        <Navbar title="My React Website" />
10
11        <Hero
12          heading="Welcome to React"
13          description="This is a simple landing page built using React components and props."
14        />
15
16        <Footer copyright="© 2026 My React Website. All Rights Reserved." />
17      </>
18    );
19  }
20
21  export default App;
```

Screenshots:

